Advances in Cognitive Systems (2026) 1-19 Submitted 9/2026

Discrimination Nets in Cognitive Science

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Abstract

Discrimination nets are tree- or graph-structured memories that recognise an input by applying a learned sequence of tests and routing the input towards a stored representation. They were central to some of the earliest computational theories of human learning, most notably the EPAM family, but they now occupy a marginal position in cognitive science. This article reviews their motivations, mechanisms, empirical coverage, and prospects. We first explain how discrimination-net models combine incremental learning, selective attention, limited short-term memory, and rapid indexing of long-term memory. We then revisit Barsalou and Bower's influential criticisms and Feigenbaum and Simon's reply, emphasising the importance of n-ary branching, recursive representations, stored images, and redundant access paths. A genealogy traces developments from EPAM through SAL, EPAM IV-VI, CHREST, MOSAIC, and CIPAL. Across these models, discrimination nets have addressed verbal learning, reading, concept formation, expert memory, chess, problem solving, language acquisition, vocabulary development, and statistical learning. We compare this tradition with neural networks and deep learning, and argue that discrimination nets remain valuable as transparent, incremental, resource-bounded models. Their strongest future role is likely to be within hybrid systems that combine learned perceptual front ends, graph-structured semantic memory, probabilistic routing, and explicit cognitive constraints.

1. Introduction

Discrimination nets were among the first data structures used in implemented theories of human cognition. Their best-known embodiment, EPAM (Elementary Perceiver and Memorizer), was developed by Edward Feigenbaum and Herbert Simon from the late 1950s onwards to explain learning and recognition in paired-associate tasks. EPAM was unusual in several respects. It was not merely a verbal description of memory, but an executable theory; it learned incrementally from the same trial sequences presented to human participants; and it connected the growth of long-term memory to assumptions about attention, processing time, and a sharply limited short-term memory (Feigenbaum, 1959; Feigenbaum & Simon, 1962a, 1984).

The discrimination-net tradition later expanded well beyond rote verbal learning. EPAM variants modelled reading, context effects in letter perception, concept attainment, classification learning, and expert memory. In the latter domain, the discrimination net was proposed as the mechanism through which experts acquire and retrieve *chunks* – that is, frequently recurring

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perceptual patterns. The related CHREST architecture applied similar mechanisms to chess perception and memory, problem solving, diagrammatic reasoning, language acquisition, vocabulary learning, and concept formation (Gobet, 2001; Gobet et al., 2001). More recent descendants, including MOSAIC and CIPAL, have used incremental chunk learning to explain crosslinguistic patterns in children's morphosyntactic errors and the discovery of words in continuous speech (Pine et al., 2023; Freudenthal et al., 2024; Jessop et al., 2025).

Despite this range, discrimination nets are now less prominent than production systems, exemplar models, and neural networks. Part of the reason is historical. The decline of classical verbal-learning research removed the empirical setting in which EPAM had been most visible. Part also reflects influential criticism, particularly Barsalou and Bower's (1984) claim that discrimination nets were psychologically implausible because they relied on test-contingent trees. In addition, the success of connectionism and, more recently, deep learning shifted attention towards distributed representations and gradient-based learning. Yet several contemporary concerns – continual learning, interpretability, resource-bounded cognition, sample efficiency, and structured memory – closely match the problems discrimination nets were designed to address.

This article therefore reassesses discrimination nets as a family of cognitive models rather than as a historical curiosity. We first define the formalism and its motivation, then present its psychologically useful features in simple terms. We next summarise the main criticisms and replies, trace the family of models descended from EPAM, and review their empirical domains. The discussion compares discrimination nets with other representations of chunks and with neural networks, before considering limitations, hybrid extensions, and future research.

1.1. Definition of a discrimination net

A discrimination net is an indexing structure that maps an input pattern onto an internal representation. Although such structures can be constructed in different ways, cognitive models such as EPAM and CHREST assume that they are acquired incrementally through experience. The distinction between the representation and its learning mechanisms is important: the former specifies how knowledge is organised and accessed, whereas the latter specifies how this organisation emerges through learning.

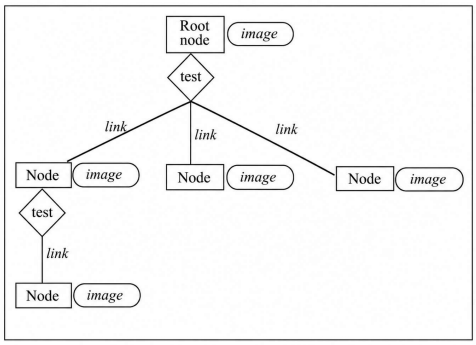


Figure 1. A discrimination net.

In its simplest form, a discrimination net is a rooted tree (see Figure 1). Internal nodes contain tests; outgoing branches represent possible outcomes of those tests; and terminal nodes contain stored descriptions, traditionally called "images". To recognise an input, the system begins at the root, applies the test at each encountered node, follows the corresponding branch, and stops at a terminal node. The terminal image is then compared with the input. A sufficiently close match constitutes recognition; a mismatch supplies information for learning (Feigenbaum & Simon, 1984).

Two complementary learning mechanisms are central to EPAM and some of its descendants. *Discrimination* occurs when an input has been routed to an inappropriate image. The system identifies a feature that distinguishes the new input from the stored image, inserts a test for that feature, and creates new branches and nodes. *Familiarisation* occurs when the input is correctly routed but the terminal image is incomplete. The system adds some of the missing information to that image. Discrimination expands the index; familiarisation enriches what is stored at its destinations. Both mechanisms are local, incremental, and interleaved with performance.

EPAM's representation is recursive. An object may contain properties and ordered subobjects, each represented in the same way as the whole. A test can therefore concern a simple property, such as colour, or the identity of a structured subobject, such as the first letter of a word. Recognition of a subobject invokes the same discrimination process, allowing a single learned letter network to be reused whenever letter identity is tested. This recursion is important because it prevents the architecture from duplicating complete subnetworks at every point where a familiar component occurs.

Discrimination nets can be binary or n-ary. EPAM II used binary tests with yes/no outcomes. EPAM III replaced these with n-ary switches, so that a test such as "What is the letter in position 1?" can have a branch for each possible value, plus branches for missing or not-yet-classified values. N-ary branching makes networks shallower and greatly weakens criticisms based on long chains of negative tests (Feigenbaum & Simon, 1984). Binary trees remain useful when the natural test is dichotomous, but they are not intrinsic to the formalism.

The formalism also admits probabilistic and weighted variants. Hintzman's (1968) SAL model attached probabilities to possible responses and allowed several responses to be associated with a stimulus, thereby generating population-level distributions without requiring every run to be deterministic. Other variants can make node creation probabilistic, represent attentional salience as the probability that a feature is selected, or attach weights such as frequency, activation, confidence, or processing time to nodes and links. In CIPAL, for example, repeated use of a chunk reduces its processing time, so frequency is expressed not merely as a count but as a changing cost of recognition (Jessop et al., 2025).

Tests may also be explicit or implicit. An explicit test queries a represented feature, such as what is the first letter of a word. An implicit test simply queries what is the next piece of information. Thus, in CHREST and MOSAIC, the descendants of a node encode what has followed a pattern in previous experience.

Finally, although the classical structure is a tree, many models are better described as networks or directed graphs. Redundant recognition paths can give a stored pattern more than one parent. Lateral links can express similarity, equivalence, production, or semantic relations. Templates add variable slots to frequently encountered structures. The tree remains the indexing backbone, but it can be embedded in a richer graph of long-term knowledge (Gobet, 1996, 2001).

1.2. Motivation behind discrimination nets

Discrimination nets provide a computational implementation of chunking, a central concept in cognitive psychology (Miller, 1956; Simon, 1974; Gobet et al., 2001). A chunk is a familiar configuration of information that can be recognised and processed as a unit. For example, an expert chess player may recognise a configuration of several pieces as a single chunk rather than processing each piece independently (Chase & Simon, 1973). Discrimination nets explain how such chunks are acquired through experience and how they are subsequently recognised and retrieved from long-term memory.

Feigenbaum and Simon (1984) described EPAM's discrimination net as an index to an encyclopaedia. The encyclopaedia is semantic long-term memory; the net is the learned index that rapidly identifies relevant entries. This distinction between index and content addresses a basic problem of expertise: as knowledge increases, why does access not become slower? A well-organised discrimination net directs processing through a small number of informative tests, so the growth of knowledge can improve rather than impair retrieval.

The motivation is also psychological economy. Humans do not store a complete, perfectly organised copy of every experience. They attend selectively, learn only part of a stimulus, and add information gradually. EPAM formalised this idea by combining slow long-term learning with limited short-term memory and an attentional noticing order. A learner initially records just enough information to distinguish an item from what is already known. Additional detail is added only when experience requires it. The resulting representation is economical in the sense of being adapted to the discrimination demands actually encountered.

This approach embodies the concept of bounded rationality (Simon, 1947). Recognition must occur in real time, with limited attention and memory, so the system cannot compare every input against every stored item. A sequential index reduces the effective search problem. It also makes learning cumulative: new chunks are built from familiar components, and later structures depend on earlier ones. The theory therefore links moment-to-moment perception to long-term development.

A further motivation is methodological. Because the model is implemented, apparently innocuous assumptions about timing, memory capacity, feature order, and learning rules have observable consequences. The theory predicts not only whether an item is recalled or classified, but also error types, learning curves, response latencies, and the internal structures produced by different learning histories. This precision was a central attraction of EPAM and remains one of the main virtues of discrimination-net models.

1.3. Features useful for simulating human behaviour

Although discrimination nets can be used as general-purpose indexing structures, EPAM and its descendants incorporate additional assumptions about human cognition. In particular, they assume that learning is incremental, that knowledge acquisition depends on selective attention and prior experience, that short-term memory is limited, and that cognitive operations take time. These assumptions make discrimination-net models more than classification algorithms: they provide process-level accounts of human learning and performance. Several features make discrimination-net models particularly valuable for explaining human cognition.

First, EPAM and CHREST acquire and modify their discrimination nets incrementally. Each experience can modify the network immediately, without retraining on an entire dataset. This fits developmental and skill-learning situations in which knowledge changes continuously.

Second, they are history-sensitive. Two learners exposed to different examples, or to the same examples in different orders, may build different nets (Gobet & Lane, 2007). Such variability is not necessarily noise; it can model individual differences and path dependence in learning.

Third, they combine recognition with memory limitations. A network is not sufficient by itself. EPAM and CHREST place retrieved chunks in a small short-term memory and assign durations to recognition, encoding, eye movements, and association. These constraints turn a data structure into a process model.

Fourth, the learned representations are inspectable. One can identify which tests route an input, which features were learned first, and why a particular error occurred. This transparency is especially valuable when the aim is to explain human behaviour rather than simply assign inputs to categories.

Fifth, discrimination nets can learn from naturalistic material. CHREST models have been trained on master chess games, parental speech, music, literature, and other structured corpora. The resulting networks preserve regularities in the environment while filtering them through cognitive constraints.

Finally, it is easy to extend the formalism to support several kinds of knowledge. A tree can index perceptual chunks; lateral links can connect related chunks; production links can associate recognised conditions with actions; and semantic links can relate nodes through labelled relations (Gobet, 1996). The same learned nodes can therefore participate in recognition, declarative knowledge, and procedural control.

1.4. Main counter-arguments of Barsalou and Bower

Barsalou and Bower (1984) offered the most extensive critique of discrimination nets as psychological models. Their arguments can be stated simply.

They first claimed that binary trees rely too heavily on negative tests. To identify an item, the system may have to ask repeatedly whether it has one feature after another, producing long and unnatural paths.

Second, they argued that test-contingent trees are overly sensitive to the order in which properties are examined. An early uninformative test may send an input down the wrong path, and the same object may require duplicate representations when different features are noticed first.

Third, they viewed discrimination nets as brittle. Missing, noisy, or incorrect properties could prevent recognition, whereas humans often recognise degraded or incomplete objects.

Fourth, they argued that networks would grow inefficiently when knowledge spans many domains. Repeating tests and subtrees could produce excessive memory requirements, perhaps an exponential increase in nodes.

Fifth, they questioned serial processing. Human perception shows important parallel effects, whereas a tree appears to evaluate one test at a time.

Finally, they argued that the formalism was incomplete as a theory of concepts and semantic memory. A routing tree can identify an item, but recognition alone does not explain similarity, flexible categorisation, meaning, or the rich relations among concepts.

1.5. Feigenbaum and Simon's reply

Feigenbaum and Simon (1984) replied that many criticisms targeted a binary-tree version that EPAM had abandoned two decades earlier. EPAM III used n-ary branches, so a single node could sort among many values. Limited "not elsewhere classified" branches were retained for a psychologically useful reason: they let the system recognise an input at a general level while also representing that it is not a familiar individual.

Feigenbaum and Simon also emphasised the role of terminal images. When a new stimulus is confused with a familiar one, the current input is compared with the stored image. A differing feature can then be selected specifically because it is discriminative. Thus, the model does not simply accept an arbitrary test order; its learning mechanism favours features that correct actual confusions. Noticing order still matters, but it is under strategic control and is supported by empirical results such as serial-position effects.

Brittleness can be reduced through redundant paths. Overlearning may connect the same object to several routes, allowing recognition from different subsets or orders of features. This extension changes a strict tree into a graph, but it preserves the core idea of learned, selective indexing.

The alleged explosion in network size also depends on misunderstanding recursion. A single letter-recognition network can be called whenever letter identity is tested; the complete subnetwork need not be copied under every word node. Feigenbaum and Simon argued that node growth is linear in the number of discriminated objects, apart from the deliberate cost of redundant routes.

On seriality, their answer was not that all perception is serial. EPAM presupposes a parallel sensory front end that extracts features, and it can coexist with parallel semantic processes. The serial component concerns selective attention and the use of extracted features for recognition. Some phenomena require seriality, while others could be handled by traversing multiple redundant paths concurrently.

Most importantly, Feigenbaum and Simon insisted that a discrimination net is only one component of EPAM. It is an index between feature extraction and semantic memory, not a complete model of either. Criticising the tree for lacking all semantics is therefore similar to criticising a library catalogue for not containing the books. This reply does not eliminate the need for a richer theory of semantic memory, but it clarifies the intended division of labour.

2. The main discrimination-net models

The history of discrimination nets is not a single linear sequence. EPAM provides the main trunk, but parallel models introduced probabilistic response mechanisms and broader tests, while CHREST generated a family of domain-specific architectures. Figure 2 summarises the principal relationships. Solid arrows indicate direct development or strong architectural inheritance; dashed boxes and arrows indicate related systems whose mechanisms were partly incorporated.

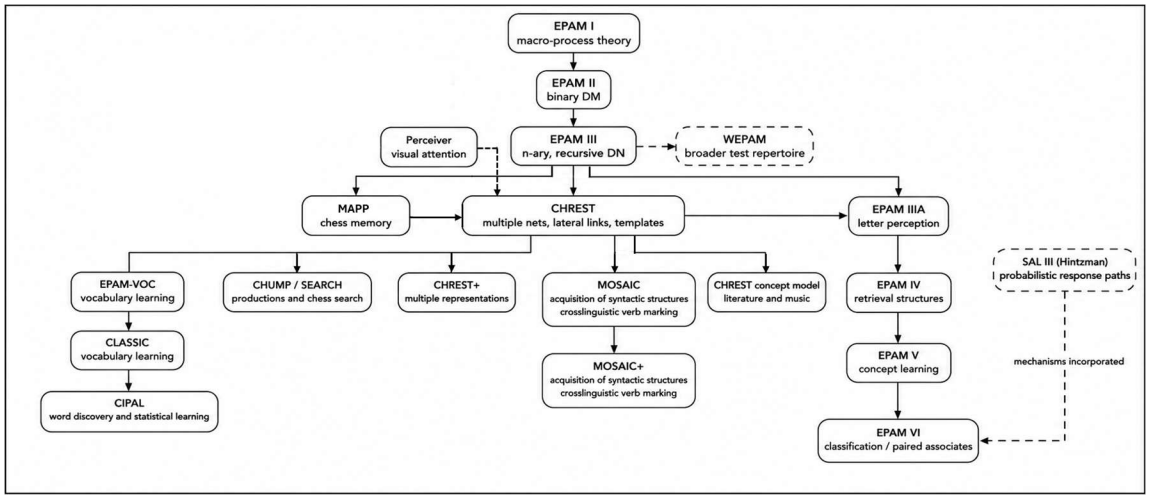


Figure 2. Genealogy of the principal EPAM-related discrimination-net models. Solid arrows indicate direct development or strong architectural inheritance; dashed boxes and arrows indicate related variants or imported mechanisms.

2.1. From EPAM I to EPAM VI

EPAM I was primarily a macro-level theory. It proposed that learning takes time, that immediate memory has very limited capacity, and that learners use anchor points to organise attention. EPAM II supplied the micro-processes and an implemented binary discrimination net. It could recognise stimuli, add tests, associate stimulus and response representations, and retrieve responses.

EPAM III (Feigenbaum & Simon, 1962b) introduced n-ary branching and a more general recursive object representation. This became the stable basis for many classical simulations of verbal learning. Its main operations were recognition, discrimination, and familiarisation, combined with a noticing strategy and limited short-term memory. Feigenbaum and Simon's 1984 article both documented this architecture and defended it against Barsalou and Bower (1984).

EPAM IIIA extended the framework to context effects in letter perception. Richman and Simon (1989) showed that a chunk-based account could reproduce the word-superiority effect while retaining serial constraints, provided that familiar words could be recognised as chunks and only a small number of chunks could be reported from a fixation.

EPAM IV added retrieval structures and activation to model the development of exceptional digit memory (Richman et al., 1995). Its network indexed semantic chunks, while a learned hierarchical retrieval structure provided slots into which digits could be rapidly encoded. The model linked years of practice to the acquisition of both semantic chunks and retrieval organisation.

EPAM V and VI broadened concept learning and paired-associate learning (Richman & Simon, 2002; Richman et al., 2002). EPAM VI incorporated ideas from SAL, including multiple possible responses and probabilistic response selection. It modelled a large set of classical verbal-learning phenomena and reproduced both the discrimination nets and latency patterns inferred from human classification data.

2.2. Parallel and related families

Hintzman's (1968) SAL III was a close relative rather than a direct EPAM version. It used discrimination-net learning for paired associates but attached probabilistic information to response

paths. SAL was important because it represented response competition and produced distributions over outcomes. Some of its mechanisms were incorporated into EPAM VI (Richman et al., 2002).

WEPAM, developed by Wynn (1966), expanded the repertoire of possible tests. Whereas standard EPAM often tested a feature at a specified position, WEPAM could test whether a feature occurred anywhere in an input. This line was not absorbed into the main architecture, but it illustrates that the psychological commitments of a discrimination net depend heavily on the available tests. Best (2006) incorporated EPAM-inspired learning mechanisms into ACT-R (Anderson & Lebière, 1998) to model production-rule induction, demonstrating how discrimination-net principles could be applied to concept learning within a different cognitive architecture.

Outside the EPAM family, decision-tree and conceptual-clustering models share structural similarities. ID3 learns trees that maximise information gain for supervised classification (Quinlan, 1986), while Cobweb incrementally constructs a probabilistic concept hierarchy (Iba & Langley, 2011; Fisher, 1987). These systems are not EPAM-like cognitive architectures: they generally lack EPAM's images, short-term-memory constraints, timing assumptions, and familiarisation mechanism. Nevertheless, they demonstrate that tree-structured learning can support both classification and concept organisation, and they provide useful algorithmic comparisons.

2.3. CHREST and its descendants

CHREST (Chunk Hierarchy and Retrieval Structures) retained EPAM's discrimination and familiarisation mechanisms but made the architecture more explicitly developmental and domain general (Gobet, 1993a, 1993b; Gobet et al., 2001). It also used ideas from MAPP (Simon & Gilmartin, 1973) and Perceiver (Simon & Barenfeld, 1969), two models of chess memory and visual attention. CHREST combined multiple modality-specific discrimination nets with short-term memories, lateral links, and templates. Similarity links connect nodes with related images; generative links use the descendants of similar nodes; production links connect conditions to actions; equivalence links connect different representations of the same entity. Templates emerge when stable patterns acquire variable slots, providing a learned form of schema (Gobet & Simon, 1996b).

Several specialised models were developed within this architecture. CHUMP (Gobet & Jansen, 1994) associated recognised chess patterns with candidate moves, whereas SEARCH (Gobet, 1997) provided a higher-level account of selective look-ahead in chess problem solving. CHREST-based models were subsequently used to explain pattern recognition and problem solving in Go (Bossomaier et al., 2012) and Awalé, an African board game (Gobet, 2009). The latter study is a rare example of computational modelling applied to cross-cultural data, in this case within developmental psychology. Finally, CHREST+ addressed the acquisition and coordination of multiple representations in physics (Lane et al., 2000).

MOSAIC (Pine et al., 2023; Freudenthal et al., 2007) adapted discrimination-net learning to sequences of words in child-directed speech. It learned utterance fragments from the edges of sentences and used similarity among continuation contexts to create generative links. Later versions modelled crosslinguistic variation in optional-infinitive errors, and MOSAIC+ added a defaulting mechanism based on the frequency distribution of verb forms (Freudenthal et al., 2024).

Other descendants focused on conceptual and lexical learning. Bennett et al. (2020) added a moving attention window and long-term chunk activation to classify unseen literary and musical sequences. EPAM-VOC (Jones et al. 2008), later renamed CLASSIC (Chunking Lexical and Sub-lexical Sequences in Children; Jones et al., 2014), aimed to explain lexical learning, successfully accounting for several phenomena in the nonword repetition task. Drawing on these models, CIPAL (Jessop et al., 2025), learns phonemic, sublexical, lexical, and multiword chunks. Each chunk has

an experience-dependent processing time, which interacts with incremental input and short-term-memory limits. Across naturalistic corpora and artificial languages, CIPAL models word discovery, statistical-learning effects, and developmental increases in processing speed.

Table 1. Principal discrimination-net models and their distinguishing features.

Model / period	Network and learning	Added mechanisms	Representative use
EPAM I (1956-59)	Macro-level theory; no full implemented DN.	Time-based learning, limited immediate memory, anchor-point attention.	Serial position and organisation of paired-associate learning.
EPAM II / III	Binary DN in EPAM II; n-ary recursive DN in EPAM III; discrimination + familiarisation.	Images, object/subobject tests, short-term memory, noticing strategies.	Verbal learning, reading, recognition, similarity and meaningfulness effects.
SAL III / WEPAM	Related DN variants; probabilistic paths in SAL; broader test set in WEPAM.	Response competition, population distributions, tests not tied to a fixed position.	Paired associates and alternative accounts of generalisation.
EPAM IIIA	EPAM III recognition net applied to visual words and letters.	Visual masking, activation, two-chunk reporting limit per fixation.	Context and word-superiority effects in letter perception.
EPAM IV	DN indexing semantic chunks plus learned retrieval structures.	Activation, episodic pointers, slotted hierarchical retrieval structures.	Acquisition of exceptional memory for long digit sequences.
EPAM V / VI	Extended DN learning for classification and paired associates.	Hypothesis tests, multiple responses, probabilistic selection, SAL mechanisms.	Concept attainment, family resemblance, latency, recognition and >40 verbal-learning effects.
CHREST	Multiple modality-specific DNs; continuous discrimination and familiarisation.	Lateral similarity / generative / production / equivalence links; templates; attention and timing.	Chess, expert memory, language, implicit learning, diagrams, development and problem solving.
CHUMP / SEARCH / CHREST+	Task-specific extensions of CHREST.	Pattern-to-move productions, selective look-ahead, links among representations.	Chess move selection and search; diagrammatic reasoning in physics.
MOSAIC / MOSAIC+	Sequential DN over words and phrases, learned from utterance edges.	Generative links from shared continuation contexts; frequency-based defaulting.	Crosslinguistic syntax development, optional-infinitive and verb-marking errors.
CHREST concept model	Sequential chunks associated with category labels.	Moving attention window and long-term chunk activation.	Classification of unseen literature and musical scores.
EPAM-VOC / CLASSIC / CIPAL	Hierarchical phonemic-to-multiword chunks; local incremental growth.	Chunk-specific processing times, memory decay, multi-level recoding.	Vocabulary, word discovery, statistical learning and processing speed.

Note. DN = discrimination net. The table groups closely related versions to save space; individual implementations sometimes differ in details.

3. Domains covered by discrimination nets

The breadth of discrimination-net research is best understood by considering what the models explain rather than by counting architectures. Table 2 summarises representative domains, references, and uses.

Verbal learning remains the foundational domain. EPAM accounted for serial-position effects, one-trial and incremental learning, forgetting, oscillation, retroactive interference, and effects of similarity, familiarity, and meaningfulness (Feigenbaum & Simon, 1962a; Simon & Feigenbaum, 1964; Gregg & Simon, 1967; Hintzman, 1967). Later EPAM VI simulations covered more than forty regularities from paired-associate research (Richman et al., 2002). The importance of this work lies not only in the number of effects but in the use of a common set of mechanisms across experiments.

Reading and letter perception extended the theory from arbitrary syllables to structured visual material. Earlier EPAM models addressed written-word recognition (Feigenbaum & Simon, 1963),

while Simon and Simon (1973) developed an EPAM-inspired generate-and-test model that used partial word-recognition information to simulate children's spelling errors. EPAM IIIA subsequently simulated the advantage for letters presented in familiar words (Richman & Simon, 1989). The model explained the effect through chunk recognition, short-term-memory limits, and visual masking rather than through unrestricted parallel activation.

Concept formation and classification provided a more demanding test. A discrimination net can represent rules as high-level tests and exceptions as lower branches. EPAM variants reproduced family-resemblance learning, old-new recognition, categorisation latencies, and the distinct networks inferred from individual participants (Richman, 1991; Gobet et al., 1997; Richman & Simon, 2002). The models also treated hypotheses as tentative tests held in short-term memory and retained only when supported by experience. CHREST has also simulated implicit learning in artificial grammar tasks, using the same discrimination and familiarisation mechanisms employed in models of chess expertise (Lane & Gobet, 2012a).

Expert memory is the second major domain. In chess, early simulations with MAPP (Simon & Gilmartin, 1973) were expanded on with CHREST (Gobet & Simon, 1996a, 1996b, 2000). CHREST learns thousands of recurring piece patterns from master games, with networks containing hundreds of thousands of chunks in simulations of masters. When presented with a chess position, the model uses its discrimination net to recognise familiar configurations of pieces. These configurations are encoded as chunks in short-term memory, allowing experts to remember more pieces than beginners despite having similarly limited short-term-memory capacity.

These chunks explained skill differences in recall, eye movements, presentation-time effects, and the residual advantage of experts even for random positions. Templates provided a mechanism for storing variable information in familiar configurations and for recalling several boards. EPAM IV applied related principles to a trained mnemonist who increased his digit span beyond one hundred digits by acquiring semantic chunks and retrieval structures (Richman et al., 1995).

Problem solving was modelled by linking recognised patterns to actions and search control. CHUMP (Gobet & Jansen, 1994) used production links from chess configurations to moves. SEARCH (Gobet, 1997) related the number of chunks and templates to the depth and speed of look-ahead. These models formalised the proposal that expertise reduces search by making relevant actions and evaluations accessible through recognition.

CHREST also addressed learning across representations. In physics (Lane et al., 2000), CHREST+ modelled how learners coordinate diagrams and equations. Other simulations treated memory for computer programs (Gobet & Oliver, 2016) and the Piagetian balance-scale task (Gobet, 1999). These applications illustrate the value of a common architecture whose input primitives and attention policies can be adapted to different domains while preserving the same learning processes.

First language acquisition became the largest post-chess application. MOSAIC (e.g., Freudenthal et al., 2007) showed how distributional learning, edge-based attention, and limited processing can produce child-like utterances and characteristic verb-marking errors without built-in adult grammatical categories. The model accounted for differences in verb-marking errors in English, Dutch, German, and Spanish. Its errors arise from fragments and combinations licensed by the statistical structure of child-directed speech. MOSAIC+ extends this account to developmental language disorder (Pine et al., 2023; Freudenthal et al., 2024).

Vocabulary development and statistical learning provide a related but distinct domain. EPAM-VOC/CLASSIC (Jones et al. 2008, Jones et al., 2014) linked phonological short-term memory to the acquisition of novel sound patterns and accounted for multiple phenomena in the nonword repetition task, a task commonly used in developmental psychology both for research and clinical practice purposes. CIPAL (Jessop et al., 2025) now offers a unified account in which larger chunks

are constructed from smaller ones, frequent chunks become faster to process, and learning is constrained by the interaction between processing time and memory decay. The architecture learned words from seventy corpora across fifteen languages and reproduced several classic artificial-language findings

Finally, discrimination nets have been applied to complex natural categories. Bennett et al. (2020) trained a CHREST-based model on literature and musical scores. It learned to classify previously unseen works by authors and composers, demonstrating that the formalism can discover sequential regularities without requiring researchers to predefine a small set of psychological dimensions. This line also illustrates a route from chunk learning to concept learning: categories emerge from distributions of stored patterns and their associations with labels.

Table 2. Representative domains modelled with discrimination nets.

Domain	Representative references	Model and use
Paired-associate and verbal learning	Feigenbaum & Simon (1962a; Simon & Feigenbaum, 1964); Gregg & Simon (1967); Richman et al. (2002)	EPAM II/III/VI: serial position, interference, oscillation, similarity, meaningfulness, one-trial and incremental learning.
Reading and letter perception	Feigenbaum & Simon (1963); Richman & Simon (1989); Simon & Simon (1973)	EPAM / EPAM IIIA: learning print-sound mappings and simulating word-superiority and context effects; simulation of spelling errors
Concept formation and implicit learning	Richman (1991); Gobet et al. (1997); Richman & Simon (2002); Lane & Gobet, 2012, 2012b	EPAM V/VI: rules, exceptions, hypotheses, family resemblance, recognition and categorisation latencies. CHREST: categorization of chess positions; implicit learning
Expert digit memory	Richman et al. (1995)	EPAM IV: semantic chunks, episodic pointers and retrieval structures acquired through extended practice.
Chess perception and memory	Simon & Gilmartin (1973); Gobet & Simon (1996a, 1996b, 2000)	MAPP / CHREST: learned piece patterns, eye movements, random-position recall, templates and presentation-time effects.
Chess problem solving	Gobet & Jansen (1994); Gobet (1997)	CHUMP / SEARCH: production links from patterns to moves and chunk-based control of selective search.
Multiple representations and other expertise	Lane et al. (2000); Gobet (2001)	CHREST+: links between diagrams and equations; related work on computer-program memory.
Cognitive development	Gobet (1999)	CHREST variant: stagewise performance in the balance-scale task from cumulative chunk learning.
Grammar and morphosyntax	Pine et al. (2023); Freudenthal et al. (2024)	MOSAIC / MOSAIC+: child-like utterances and crosslinguistic verb-marking errors from input distribution and processing constraints.
Vocabulary and word discovery	Gobet et al. (2001); Jessop et al. (2025)	CLASSIC / CIPAL: phonological chunks, nonword repetition, word segmentation, statistical learning and developmental speed-up.
Natural sequential concepts	Bennett et al. (2020)	CHREST extension: learning and classifying authorial and musical styles from raw symbolic sequences.

Note. The table is selective rather than exhaustive and emphasises simulations that use EPAM/CHREST-family learning mechanisms.

4. Discussion

4.1. Other methods for representing chunks

Discrimination nets are only one way to represent chunks. Production systems represent knowledge as condition-action rules. In ACT-R (Anderson & Lebière, 1998) and Soar (Newell, 1990; Laird, 2012), a familiar configuration may be encoded as a production whose condition matches working memory and whose action retrieves information or changes the state of the system. Chunking in

Soar refers primarily to compiling the result of problem solving into a new production, which is more goal-oriented than the continuous perceptual chunking of EPAM and CHREST.

Semantic networks represent chunks or concepts as nodes connected by labelled relations. They are well suited to taxonomic, partonomic, causal, and associative knowledge but require a separate process for rapidly recognising which node an input instantiates. Gobet's (1996) composite framework treated the discrimination net as the index and semantic links as the richer knowledge reached through that index. This division remains attractive because it separates perceptual access from semantic content without treating them as independent.

Hierarchical category models provide another way of organizing learned representations. Cobweb (Fisher, 1987; Iba & Langley, 2011) constructs a probabilistic concept hierarchy incrementally from examples, without requiring predefined category labels. Each node represents a category through the probability distributions of its attributes, and the organization of the hierarchy is guided by category utility, a measure of how informative the categories are. Cobweb can also merge or split categories as new information becomes available, while subsequent work has explored more extensive methods for revising an existing hierarchy (Fisher, 1996). Like EPAM, Cobweb combines recognition with the gradual organization of experience. However, it relies on probabilistic category descriptions rather than learned sequences of feature tests leading to stored images, and it does not incorporate EPAM's detailed assumptions about attention, memory limitations, and processing time. The comparison illustrates that similar hierarchical knowledge structures can be constructed and modified through different learning mechanisms.

Exemplar and prototype models represent categories through stored cases or central tendencies (Kruschke, 1992; Langley, 2025). They often provide flexible similarity-based classification and account well for graded typicality. However, their psychological implementations may say less about how perceptual units are constructed from raw sequences, how memory access unfolds over time, or how a large set of exemplars is indexed. Discrimination nets can complement these accounts by learning the units and retrieval paths on which similarity computations operate.

Neural networks represent chunks as distributed activation patterns and encode regularities in connection weights. This offers smooth generalisation, tolerance of noise, and learned feature extraction. Recurrent and convolutional networks can discover local patterns that function like chunks, although the resulting representations are not normally discrete or directly inspectable. Adaptive Resonance Theory, self-organising maps, and staged recurrent learning are examples in which clustering or local dependencies have an explicitly chunk-like interpretation (Gobet et al., 2001).

4.2. Discrimination nets versus neural networks and deep learning

The contrast between discrimination nets and neural networks is often overstated. Both focus on perception as the primary driver of intelligence/cognition; both learn from environmental regularities; both alter retrieval as a function of experience; and both can implement hierarchical recognition. The main differences concern representation, learning signals, and allocation of computational resources.

A classical discrimination net stores discrete tests, branches, and images. Its effective parameters are the nodes, links, test choices, branch values, timing constants, and any frequencies or probabilities attached to them. A large CHREST network may contain hundreds of thousands of nodes, but many timing assumptions are shared globally and learning adds structures only when experience requires them. Deep neural networks, by contrast, typically contain large arrays of continuous weights trained together to minimise a global objective. Their capacity is often distributed across the network rather than associated with individually interpretable memories.

The contrast is easier to state than to quantify, and the tradition has not made its own case as forcefully as it might. The relevant number is not how many structures a model stores but how many quantities the modeller is free to set. In the chess simulations of CHREST, the learned network is determined almost entirely by the training corpus and the order in which it is presented; what the modeller supplies is roughly ten architectural constants – 8-10 s to create a node, 2 s to add information to an existing image, 10 ms to traverse a link, 50 ms to place a recognised chunk in short-term memory, 250 ms to fill a template slot, a visual short-term memory of three chunks, a visual field of two squares around the fixation point, and the thresholds governing template formation and similarity links (Gobet, 1993b; De Groot et al., 1996; Gobet & Simon, 2000). These values were carried over from earlier EPAM work rather than re-estimated for the simulations, or were estimated based on independent studies (Zhang & Simon, 1985; Richman et al., 1995). CIPAL states the same discipline outright. Its timing parameters – a speech rate of 160 ms per phoneme calculated from child-directed corpora, an 800 ms decay taken from the mismatch-negativity literature, and a 1,200 ms initial processing time anchored to looking-while-listening latencies – were kept at their default values and were identical for every simulation across seventy corpora in fifteen languages (Jessop et al., 2025). In deep neural networks, by contrast, connection weights are typically estimated from training data, while architectural parameters and training hyperparameters are specified separately. The relevant comparison is therefore not simply the number of parameters, but the extent to which a model's assumptions are independently constrained by psychological evidence rather than adjusted to fit the data. This accounting deserves to be reported explicitly, model by model. Covering the phenomena of Table 2 with a handful of constants fixed in advance is a far stronger claim than covering them with parameters re-tuned per phenomenon, and a reader cannot currently tell from the literature which claim holds.

This difference has consequences for sample efficiency and forgetting. EPAM-style learning is local and incremental: a new discrimination is added at the point of confusion, while unrelated parts of memory remain unchanged. Standard backpropagation modifies many weights (Rumelhart et al., 1986) and can interfere with prior knowledge unless rehearsal, regularisation, or architectural protection is used. However, neural networks generally learn perceptual features that discrimination nets take as primitives, and they interpolate smoothly in high-dimensional spaces where symbolic tests may be difficult to specify.

Inference also differs. A tree uses conditional computation: only the tests on one or a few paths are evaluated. A dense neural network normally activates much of its structure for each input. The tree therefore provides an explicit trace of the decision and can be computationally economical at inference. Neural networks can be faster on parallel hardware and more robust to distributed noise, but their reasoning path is less transparent.

Can a discrimination net emulate backpropagation? Not in its classical psychological form. Discrimination and familiarisation are local symbolic operations driven by mismatches, whereas backpropagation computes gradients through a differentiable function with respect to a global loss. In a strict sense, sufficiently enriched symbolic systems can simulate any computable learning procedure, but this does not make gradient descent a natural explanation at the cognitive level. More productive is to ask whether tree-structured routing can be made differentiable. Deep neural decision forests use stochastic, differentiable splits so that routing parameters and perceptual representations can be trained by backpropagation (Kontschieder et al., 2015). Adaptive neural trees go further by placing learned neural transformations on branches and leaves while growing the tree structure (Tanno et al., 2019). These systems show that trees and backpropagation are technically compatible, although they do not inherit EPAM's assumptions about human attention, images, familiarisation, or processing time.

Three developments in large language models bear more directly on the tradition than the tree-structured networks just discussed. Subword tokenisation builds a hierarchy of reusable units by repeatedly merging the most frequent adjacent pair in a corpus (Gage, 1994; Sennrich et al., 2016). This is chunking by frequency, in Miller's (1956) original sense of recoding the input into progressively larger units, and it proceeds by the same operation – concatenation of adjacent units into a growing inventory – through which CLASSIC and CIPAL build their chunk hierarchies. The instructive difference lies in what has been removed. Byte-pair encoding is an offline computation over whole-corpus statistics – no memory limit, no processing time, no interleaving of learning with performance, and a single free parameter, the number of merges, set by engineering choice rather than by psychology. It is a chunking algorithm that lacks the psychologically motivated memory and processing-time constraints central to EPAM and CHREST (see Section 1.3).[1] The connection is now being drawn within natural language processing itself, where cognitively inspired tokenisers appeal explicitly to chunking and least-effort principles (Yang, 2024). Sparse mixture-of-experts architectures route each input to a small subset of the network (Shazeer et al., 2017; Fedus et al., 2022), so conditional computation no longer distinguishes trees; what still distinguishes them is that the routing function is learned against a global objective, and the resulting route, though loggable, is not a sequence of interpretable tests – analyses of trained routers find little human-interpretable specialisation (Jiang et al., 2024). Finally, retrieval-augmented generation (RAG) separates an index from the content it retrieves (Lewis et al., 2020), echoing the division of labour proposed by Feigenbaum and Simon. The use of a similar architecture in modern AI systems illustrates the generality of this approach to memory organisation and retrieval. Nevertheless, the analogy is limited: RAG systems typically retrieve information through embedding similarity rather than a learned sequence of symbolic tests, and their learning mechanisms do not incorporate EPAM's assumptions about incremental discrimination, familiarisation, attention, and processing time.

The reverse emulation is also possible: a neural network can approximate the input-output behaviour of a discrimination tree, and a tree can be distilled from a trained network. Yet behavioural equivalence does not imply theoretical equivalence. For cognitive science, the relevant question is which representation and learning process explain human errors, timing, transfer, and development with the fewest unsupported assumptions.

The tree-versus-graph distinction is similarly graded. EPAM III is a tree because each image has one principal access path. CHREST adds redundant paths and lateral links, turning long-term memory into a directed graph. Neural networks are also graphs, but their edges usually carry continuously valued weights and their nodes perform uniform differentiable computations. A graph-based discrimination system could occupy an intermediate position: discrete chunks and interpretable links, but probabilistic routing, activation spreading, and multiple simultaneous paths.

4.3. Limitations and possible extensions

The first limitation is the choice of primitives. Most discrimination-net models assume that an input has already been encoded into symbolic features: letters, phonemes, words, chess pieces, or spatial relations. This assumption is reasonable when the research question concerns learning above

[1] The sentence compresses two claims of different standing. The identification of byte-pair encoding as chunking rests on Miller's (1956) compression sense of the term and on the shared concatenation operation noted in the text. It is scoped by "chunking by frequency", since merging is driven by raw pair frequency, whereas a chunk is defined relationally, by strong associations within and weak associations without (Gobet et al., 2001). The absences are definitional rather than incidental – the algorithm is specified over global corpus counts, and its vocabulary is fixed before use (Sennrich et al., 2016).

perception, but it leaves open how those primitives are extracted. A neural perceptual front end could map images, speech, or sensor data onto candidate symbolic features, with the discrimination net learning stable chunks over that representation.

Second, classical networks have limited semantics. An image stores a description and associations can retrieve other nodes, but a tree alone does not represent rich relational knowledge, compositional meaning, or causal structure. The solution proposed in the 1996 composite framework remains relevant: use the discrimination net as an index into a semantic graph and allow the same nodes to participate in productions and labelled relations. Modern knowledge graphs and vector embeddings could supply additional semantic content, provided that their role is specified at the process level.

Third, exact symbolic tests can generalise poorly when categories vary continuously. Probabilistic branches, partial matching, ranges, learned thresholds, and prototype-like images offer possible remedies. SAL demonstrated probabilistic response paths; EPAM VI treated feature salience probabilistically; CHREST uses similarity to form lateral links; and CIPAL places graded processing costs on chunks. A future architecture could combine these mechanisms within a unified Bayesian or activation-based routing scheme while preserving inspectable structure.

Fourth, the input requirements can be demanding. Because the network is built from experience, simulations need representative training material in the correct order and at an appropriate scale. This is a theoretical strength when environmental exposure is known, but a practical weakness when developmental input is unavailable. Corpus uncertainty should therefore be treated explicitly, with sensitivity analyses over plausible exposure histories.

Fifth, large networks are difficult to analyse. A chess model with hundreds of thousands of partially redundant chunks may be more psychologically plausible than a compact hand-written rule set, but transparency at the level of an individual node does not guarantee transparency of the whole system. Tools are needed for summarising network motifs, tracing the provenance of chunks, measuring redundancy, and comparing the structures learned by different simulated individuals.

Sixth, most models simplify forgetting, consolidation, and neural implementation. Some EPAM versions include activation, decay, and loss from retrieval structures, while CIPAL changes processing time with use. A fuller theory would separate short-lived activation, long-term strengthening, interference, and structural deletion. It would also link symbolic chunks to candidate neural mechanisms. Recent work comparing CHREST-like chunks with learned neural representations (Bennett & Gobet, 2024) suggests one possible bridge, but the mapping between cognitive and neural levels remains underdeveloped.

These limitations suggest two research avenues. The first involves developing a hybrid architecture that combines different representational and learning mechanisms. A future discrimination architecture could use neural networks for feature extraction, a probabilistic discrimination graph for incremental indexing, a semantic graph for relational knowledge, and production or planning mechanisms for action. Learning could combine local symbolic growth with gradient-based tuning of perceptual and routing functions. Crucially, the architecture should retain explicit processing times, limited working memory, and trial-by-trial learning if it is intended as a cognitive model rather than merely an engineering classifier.

The second avenue is more parsimonious and aligns with the *Single Algorithm Hypothesis* in neuroscience, which proposes that visual, auditory, motor, and somatosensory brain regions rely on broadly similar computational principles to extract comparable structural representations from different types of information (Hawkins & Blakeslee, 2004). A related idea appears in artificial general intelligence (AGI), where a single learning algorithm is expected to exhibit competence across a wide range of tasks and domains (Silver et al., 2017). The central question is whether human cognition can be explained in terms of chunking mechanisms similar to those underlying

EPAM and CHREST, and, in particular, whether these mechanisms can be generalised to account for the emergence and representation of semantic knowledge.

4.4. Do discrimination nets have a future in cognitive science?

Discrimination nets are unlikely to displace deep learning as general-purpose perceptual technology. They should instead be judged against the goals of cognitive science: explaining how experience produces structured knowledge under constraints, how that knowledge is accessed, and why people show particular patterns of success and error.

On these criteria, the formalism retains several advantages. It provides an explicit model of memory indexing; it learns continuously without requiring global retraining; it produces inspectable histories and decision paths; it naturally supports conditional computation; and it connects environmental structure to the growth of chunks. The empirical record is unusually broad for a family based on a small set of mechanisms. Verbal learning, expert memory, classification, and language acquisition are not superficial demonstrations but domains in which detailed behavioural patterns have been simulated.

The most promising future lies in four directions. First, discrimination nets can serve as models of developmental and continual learning, especially when order of experience and resource limits matter. Second, they can provide structured, interpretable memory components within neuro-symbolic systems. Third, graph extensions can integrate recognition with semantic and procedural knowledge. Fourth, automated model discovery can explore variants of tests, attention policies, and learning rules while preserving the constraints and transparency of the architecture.

The tradition also offers a methodological warning. A model should not be rejected because a simplified data structure appears intuitively implausible, nor accepted because it fits aggregate accuracy. The relevant unit of evaluation is the complete process theory: representation, learning history, attention, memory limits, timing, and output. Feigenbaum and Simon's defence of EPAM made this point forty years ago, and it is even more important in an era of highly parameterised models.

Discrimination nets therefore have a future, but probably not as isolated trees. Their enduring contribution is the idea that cognition requires a learned index: a structure that selects, compresses, and organises experience so that a bounded system can reach relevant knowledge quickly. Combined with modern perceptual learning and richer semantic representations, this idea remains a strong candidate component of human-like cognitive systems.

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